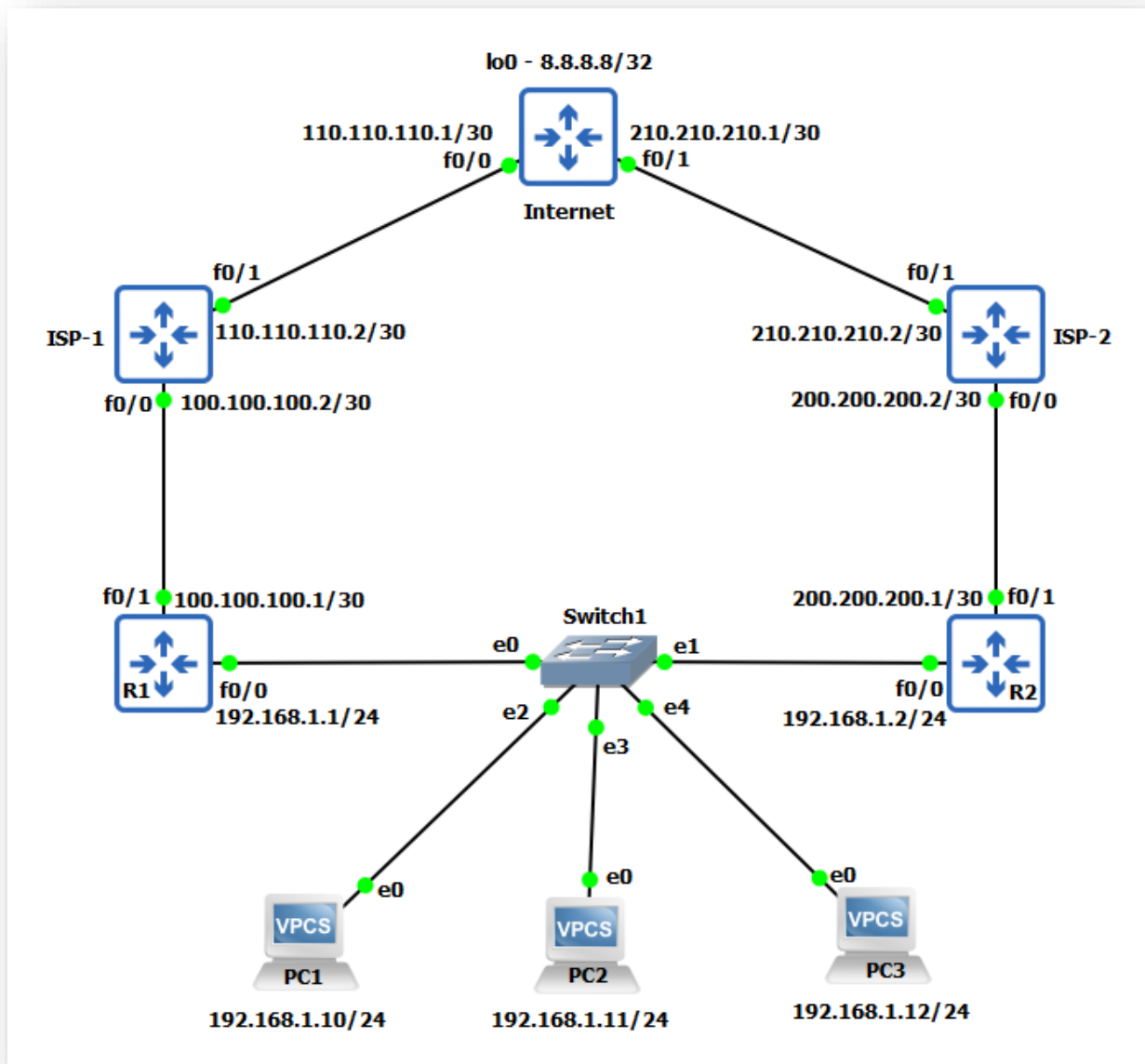


Virtual Router Redundancy Protocol (VRRP)



Task

1. Configure routers with IP address as shown in topology and configure enable password as **ccna**. (**Refer Lab-22 for task-1 to task-5**)
2. Configure VPCs with IP address and default gateway of 192.168.1.3 (VRRP Virtual IP)
3. Configure default route on R1 & R2 pointing to ISP
4. Configure OSPF on ISP-1, ISP-2 and Internet router.
5. Configure NAT overload (PAT) on R1 & R2
6. Configure VRRP with virtual IP 192.168.1.3 such that R1 will be Master and R2 will be Backup. Enable clear text authentication with secret key **Ratan**
7. Configure IP SLA to detect indirect ISP link failure on R1.

Task-6: Configure VRRP with virtual IP 192.168.1.3 such that R1 will be Master and R2 will be Backup. Enable clear text authentication with secret key Ratan

R1 Configuration:

```
R1#config t
R1(config)#int fa0/0
R1(config-if)#vrrp 1 ip 192.168.1.3
R1(config-if)#vrrp 1 priority 105
R1(config-if)#vrrp 1 authentication text Ratan
R1(config-if)#exit
R1(config)#exit
R1#
```

R2 Configuration:

```
R2#config t
R2(config)#int fa0/0
R2(config-if)#vrrp 1 ip 192.168.1.3
R2(config-if)#vrrp 1 authentication text Ratan
R2(config-if)#exit
R2(config)#exit
R2#
```

✓ Verification & Testing:

R1#sh vrrp

```
FastEthernet0/0 - Group 1
State is Master
Virtual IP address is 192.168.1.3
Virtual MAC address is 0000.5e00.0101
Advertisement interval is 1.000 sec
Preemption enabled
Priority is 105
Authentication text, string "Ratan"
Master Router is 192.168.1.1 (local), priority is 105
Master Advertisement interval is 1.000 sec
Master Down interval is 3.589 sec
R1#
```

R1#sh vrrp brief

Interface	Grp	Pri	Time	Own	Pre	State	Master addr	Group addr
Fa0/0	1	105	3589		Y	Master	192.168.1.1	192.168.1.3

R1#

R2#sh vrrp

```
FastEthernet0/0 - Group 1
State is Backup
Virtual IP address is 192.168.1.3
Virtual MAC address is 0000.5e00.0101
Advertisement interval is 1.000 sec
```

Preemption enabled

Priority is 100

Authentication text, string "Ratan"

Master Router is 192.168.1.1, priority is 105

Master Advertisement interval is 1.000 sec

Master Down interval is 3.609 sec (expires in 2.969 sec)

R2#

R2#sh vrrp brief

Interface	Grp	Pri	Time	Own	Pre	State	Master addr	Group addr
Fa0/0	1	100	3609		Y	Backup	192.168.1.1	192.168.1.3

R2#

Ping VRRP Virtual IP 192.168.1.3 from PCs

PC1> ping 192.168.1.3

84 bytes from 192.168.1.3 icmp_seq=1 ttl=255 time=9.324 ms
 84 bytes from 192.168.1.3 icmp_seq=2 ttl=255 time=11.615 ms
 84 bytes from 192.168.1.3 icmp_seq=3 ttl=255 time=5.994 ms
 84 bytes from 192.168.1.3 icmp_seq=4 ttl=255 time=6.519 ms
 84 bytes from 192.168.1.3 icmp_seq=5 ttl=255 time=11.507 ms

PC2> ping 192.168.1.3

84 bytes from 192.168.1.3 icmp_seq=1 ttl=255 time=9.682 ms
 84 bytes from 192.168.1.3 icmp_seq=2 ttl=255 time=6.478 ms
 84 bytes from 192.168.1.3 icmp_seq=3 ttl=255 time=4.274 ms
 84 bytes from 192.168.1.3 icmp_seq=4 ttl=255 time=3.403 ms
 84 bytes from 192.168.1.3 icmp_seq=5 ttl=255 time=10.435 ms

PC3> ping 192.168.1.3

84 bytes from 192.168.1.3 icmp_seq=1 ttl=255 time=9.244 ms
 84 bytes from 192.168.1.3 icmp_seq=2 ttl=255 time=9.336 ms
 84 bytes from 192.168.1.3 icmp_seq=3 ttl=255 time=5.562 ms
 84 bytes from 192.168.1.3 icmp_seq=4 ttl=255 time=12.519 ms
 84 bytes from 192.168.1.3 icmp_seq=5 ttl=255 time=11.434 ms

Ping Internet IP 8.8.8.8 from PCs

PC1> ping 8.8.8.8

84 bytes from 8.8.8.8 icmp_seq=1 ttl=253 time=53.327 ms
 84 bytes from 8.8.8.8 icmp_seq=2 ttl=253 time=53.359 ms
 84 bytes from 8.8.8.8 icmp_seq=3 ttl=253 time=49.293 ms
 84 bytes from 8.8.8.8 icmp_seq=4 ttl=253 time=44.325 ms
 84 bytes from 8.8.8.8 icmp_seq=5 ttl=253 time=51.393 ms

PC2> ping 8.8.8.8

84 bytes from 8.8.8.8 icmp_seq=1 ttl=253 time=53.392 ms
 84 bytes from 8.8.8.8 icmp_seq=2 ttl=253 time=54.279 ms
 84 bytes from 8.8.8.8 icmp_seq=3 ttl=253 time=52.346 ms
 84 bytes from 8.8.8.8 icmp_seq=4 ttl=253 time=44.337 ms
 84 bytes from 8.8.8.8 icmp_seq=5 ttl=253 time=49.133 ms

PC3> ping 8.8.8.8

84 bytes from 8.8.8.8 icmp_seq=1 ttl=253 time=53.142 ms
 84 bytes from 8.8.8.8 icmp_seq=2 ttl=253 time=55.309 ms
 84 bytes from 8.8.8.8 icmp_seq=3 ttl=253 time=55.278 ms
 84 bytes from 8.8.8.8 icmp_seq=4 ttl=253 time=52.222 ms
 84 bytes from 8.8.8.8 icmp_seq=5 ttl=253 time=51.272 ms

R1#sh ip nat translations

Pro	Inside global	Inside local	Outside local	Outside global
icmp	100.100.100.1:5571	192.168.1.10:5571	8.8.8.8:5571	8.8.8.8:5571
icmp	100.100.100.1:5827	192.168.1.10:5827	8.8.8.8:5827	8.8.8.8:5827
icmp	100.100.100.1:6083	192.168.1.10:6083	8.8.8.8:6083	8.8.8.8:6083
icmp	100.100.100.1:6339	192.168.1.10:6339	8.8.8.8:6339	8.8.8.8:6339
icmp	100.100.100.1:6595	192.168.1.10:6595	8.8.8.8:6595	8.8.8.8:6595
icmp	100.100.100.1:7107	192.168.1.11:7107	8.8.8.8:7107	8.8.8.8:7107
icmp	100.100.100.1:7363	192.168.1.11:7363	8.8.8.8:7363	8.8.8.8:7363
icmp	100.100.100.1:7619	192.168.1.11:7619	8.8.8.8:7619	8.8.8.8:7619
icmp	100.100.100.1:7875	192.168.1.11:7875	8.8.8.8:7875	8.8.8.8:7875
icmp	100.100.100.1:8131	192.168.1.11:8131	8.8.8.8:8131	8.8.8.8:8131
icmp	100.100.100.1:8643	192.168.1.12:8643	8.8.8.8:8643	8.8.8.8:8643
icmp	100.100.100.1:8899	192.168.1.12:8899	8.8.8.8:8899	8.8.8.8:8899
icmp	100.100.100.1:9411	192.168.1.12:9411	8.8.8.8:9411	8.8.8.8:9411
icmp	100.100.100.1:9667	192.168.1.12:9667	8.8.8.8:9667	8.8.8.8:9667
icmp	100.100.100.1:9923	192.168.1.12:9923	8.8.8.8:9923	8.8.8.8:9923

R1#

R2#sh ip nat translations

R2#

PC1> trace 8.8.8.8

trace to 8.8.8.8, 8 hops max, press Ctrl+C to stop
 1 192.168.1.1 20.362 ms 9.325 ms 9.506 ms
 2 100.100.100.2 30.284 ms 32.290 ms 32.438 ms
 3 *110.110.110.1 52.202 ms (ICMP type:3, code:3, Destination port unreachable)

PC2> trace 8.8.8.8

trace to 8.8.8.8, 8 hops max, press Ctrl+C to stop
 1 192.168.1.1 3.276 ms 9.389 ms 9.539 ms
 2 100.100.100.2 32.416 ms 31.285 ms 31.277 ms
 3 *110.110.110.1 53.197 ms (ICMP type:3, code:3, Destination port unreachable)

PC3> trace 8.8.8.8

trace to 8.8.8.8, 8 hops max, press Ctrl+C to stop
 1 192.168.1.1 4.653 ms 10.539 ms 9.759 ms
 2 100.100.100.2 31.390 ms 31.397 ms 30.435 ms
 3 *110.110.110.1 52.358 ms (ICMP type:3, code:3, Destination port unreachable)

R1#sh arp

Protocol	Address	Age (min)	Hardware Addr	Type	Interface
Internet	100.100.100.1	-	ca01.25e0.0006	ARPA	FastEthernet0/1
Internet	100.100.100.2	30	ca03.09c8.0008	ARPA	FastEthernet0/1

```

Internet 192.168.1.1      -   ca01.25e0.0008  ARPA  FastEthernet0/0
Internet 192.168.1.3      -   0000.5e00.0101  ARPA  FastEthernet0/0
Internet 192.168.1.10     0   0050.7966.6800  ARPA  FastEthernet0/0
Internet 192.168.1.11     2   0050.7966.6801  ARPA  FastEthernet0/0
Internet 192.168.1.12     2   0050.7966.6802  ARPA  FastEthernet0/0
R1#

```

Test – From PC1 start the PING for 8.8.8.8 and simulate link failure by shutting down the R1's LAN interface.

```

R1#config t
R1(config)#int fa0/0
R1(config-if)#shut
R1(config)#exit
R1#

```

PC1> ping 8.8.8.8 -c 20

```

84 bytes from 8.8.8.8 icmp_seq=1 ttl=253 time=53.143 ms
84 bytes from 8.8.8.8 icmp_seq=2 ttl=253 time=44.260 ms
84 bytes from 8.8.8.8 icmp_seq=3 ttl=253 time=54.203 ms
84 bytes from 8.8.8.8 icmp_seq=4 ttl=253 time=52.327 ms
84 bytes from 8.8.8.8 icmp_seq=5 ttl=253 time=53.306 ms
8.8.8.8 icmp_seq=6 timeout
84 bytes from 8.8.8.8 icmp_seq=7 ttl=253 time=111.094 ms
84 bytes from 8.8.8.8 icmp_seq=8 ttl=253 time=47.391 ms
84 bytes from 8.8.8.8 icmp_seq=9 ttl=253 time=47.530 ms
84 bytes from 8.8.8.8 icmp_seq=10 ttl=253 time=53.324 ms
84 bytes from 8.8.8.8 icmp_seq=11 ttl=253 time=49.415 ms
84 bytes from 8.8.8.8 icmp_seq=12 ttl=253 time=47.463 ms
84 bytes from 8.8.8.8 icmp_seq=13 ttl=253 time=50.636 ms
84 bytes from 8.8.8.8 icmp_seq=14 ttl=253 time=55.437 ms
84 bytes from 8.8.8.8 icmp_seq=15 ttl=253 time=49.037 ms
84 bytes from 8.8.8.8 icmp_seq=16 ttl=253 time=52.429 ms
84 bytes from 8.8.8.8 icmp_seq=17 ttl=253 time=47.437 ms
84 bytes from 8.8.8.8 icmp_seq=18 ttl=253 time=52.178 ms
84 bytes from 8.8.8.8 icmp_seq=19 ttl=253 time=43.888 ms
84 bytes from 8.8.8.8 icmp_seq=20 ttl=253 time=53.244 ms

```

R1#

***Nov 21 19:35:45.079: %VRRP-6-STATECHANGE: Fa0/0 Grp 1 state Master -> Init**

R1#

R2#

***Nov 21 19:35:35.539: %VRRP-6-STATECHANGE: Fa0/0 Grp 1 state Backup -> Master**

R2#

R1#sh vrrp

FastEthernet0/0 - Group 1

State is Init

Virtual IP address is 192.168.1.3

Virtual MAC address is 0000.5e00.0101

Advertisement interval is 1.000 sec

Preemption enabled

Priority is 105

Authentication text, string "Ratan"

Master Router is unknown, priority is unknown

Master Advertisement interval is unknown

Master Down interval is unknown

R1#sh vrrp brief

Interface	Grp	Pri	Time	Own	Pre	State	Master addr	Group addr
Fa0/0	1	105	3589		Y	Init	0.0.0.0	192.168.1.3

R1#

R2#sh vrrp

FastEthernet0/0 - Group 1

State is Master

Virtual IP address is 192.168.1.3

Virtual MAC address is 0000.5e00.0101

Advertisement interval is 1.000 sec

Preemption enabled

Priority is 100

Authentication text, string "Ratan"

Master Router is 192.168.1.2 (local), priority is 100

Master Advertisement interval is 1.000 sec

Master Down interval is 3.609 sec

R2#sh vrrp brief

Interface	Grp	Pri	Time	Own	Pre	State	Master addr	Group addr
Fa0/0	1	100	3609		Y	Master	192.168.1.2	192.168.1.3

R2#

PC1> ping 8.8.8.8

84 bytes from 8.8.8.8 icmp_seq=1 ttl=253 time=67.211 ms
 84 bytes from 8.8.8.8 icmp_seq=2 ttl=253 time=47.141 ms
 84 bytes from 8.8.8.8 icmp_seq=3 ttl=253 time=54.335 ms
 84 bytes from 8.8.8.8 icmp_seq=4 ttl=253 time=51.309 ms
 84 bytes from 8.8.8.8 icmp_seq=5 ttl=253 time=49.451 ms

PC1> trace 8.8.8.8

trace to 8.8.8.8, 8 hops max, press Ctrl+C to stop

```

1  192.168.1.2  9.360 ms  10.436 ms  10.376 ms
2  200.200.200.2  31.312 ms  31.346 ms  31.264 ms
3  *210.210.210.1  53.224 ms (ICMP type:3, code:3, Destination port unreachable)
  
```

PC2> ping 8.8.8.8

84 bytes from 8.8.8.8 icmp_seq=1 ttl=253 time=49.562 ms
 84 bytes from 8.8.8.8 icmp_seq=2 ttl=253 time=53.218 ms
 84 bytes from 8.8.8.8 icmp_seq=3 ttl=253 time=49.276 ms
 84 bytes from 8.8.8.8 icmp_seq=4 ttl=253 time=53.257 ms
 84 bytes from 8.8.8.8 icmp_seq=5 ttl=253 time=53.219 ms

PC2> trace 8.8.8.8

trace to 8.8.8.8, 8 hops max, press Ctrl+C to stop

```
1 192.168.1.2 7.703 ms 9.603 ms 10.647 ms
2 200.200.200.2 30.302 ms 30.413 ms 30.420 ms
3 *210.210.210.1 53.249 ms (ICMP type:3, code:3, Destination port unreachable)
```

PC3> ping 8.8.8.8

```
84 bytes from 8.8.8.8 icmp_seq=1 ttl=253 time=52.128 ms
84 bytes from 8.8.8.8 icmp_seq=2 ttl=253 time=55.628 ms
84 bytes from 8.8.8.8 icmp_seq=3 ttl=253 time=45.410 ms
84 bytes from 8.8.8.8 icmp_seq=4 ttl=253 time=48.274 ms
84 bytes from 8.8.8.8 icmp_seq=5 ttl=253 time=55.263 ms
```

PC3> trace 8.8.8.8

trace to 8.8.8.8, 8 hops max, press Ctrl+C to stop

```
1 192.168.1.2 7.485 ms 9.517 ms 9.435 ms
2 200.200.200.2 32.348 ms 30.256 ms 31.366 ms
3 *210.210.210.1 52.228 ms (ICMP type:3, code:3, Destination port unreachable)
```

Test – Simulate link recovery by making R1's LAN interface up.

```
R1#config t
R1(config)#int fa0/0
R1(config-if)#no shut
R1(config-if)#exit
R1(config)#exit
R1#
```

R1#

```
*Nov 21 19:45:02.283: %VRRP-6-STATECHANGE: Fa0/0 Grp 1 state Backup -> Master
```

R1#

R2#

```
*Nov 21 19:44:52.155: %VRRP-6-STATECHANGE: Fa0/0 Grp 1 state Master -> Backup
```

R2#

R1#sh vrrp

FastEthernet0/0 - Group 1

State is Master

Virtual IP address is 192.168.1.3

Virtual MAC address is 0000.5e00.0101

Advertisement interval is 1.000 sec

Preemption enabled

Priority is 105

Authentication text, string "Ratan"

Master Router is 192.168.1.1 (local), priority is 105

Master Advertisement interval is 1.000 sec

Master Down interval is 3.589 sec

R1#

R2#sh vrrp

FastEthernet0/0 - Group 1

State is Backup

Virtual IP address is 192.168.1.3

Virtual MAC address is 0000.5e00.0101

Advertisement interval is 1.000 sec

Preemption enabled

Priority is 100

Authentication text, string "Ratan"

Master Router is 192.168.1.1, priority is 105

Master Advertisement interval is 1.000 sec

Master Down interval is 3.609 sec (expires in 3.589 sec)

Task-7: Configure IP SLA to detect indirect ISP link failure on R1.**R1 Configuration:**

```

R1#config t
R1(config)#ip sla 1
R1(config-ip-sla)#icmp-echo 8.8.8.8
R1(config-ip-sla-echo)#exit
R1(config)#ip sla schedule 1 start-time now life forever
R1(config)#track 1 ip sla 1 reachability
R1(config-track)#exit
R1(config)#int fa0/0
R1(config-if)#vrrp 1 track 1 decrement 20
R1(config-if)#exit
R1(config)#exit
R1#

```

✓ Verification & Testing:**R1#sh track**

Track 1

IP SLA 1 reachability

Reachability is Up

2 changes, last change 00:01:37

Latest operation return code: OK

Latest RTT (milliseconds) 32

Tracked by:

VRRP FastEthernet0/0 1

R1#

R1#sh ip sla summary

IPSLAs Latest Operation Summary

Codes: * active, ^ inactive, ~ pending

ID	Type	Destination	Stats (ms)	Return Code	Last Run
*1	icmp-echo	8.8.8.8	RTT=32	OK	34 seconds ago

Test – From PC1 start the PING for 8.8.8.8 and simulate link failure by shutting down the ISP-1 fa0/1 interface (indirect interface).

```
ISP-1#config t
ISP-1(config)#int fa0/1
ISP-1(config-if)#shut
ISP-1(config-if)#exit
ISP-1#
```

PC1> ping 8.8.8.8 -c 20

```
84 bytes from 8.8.8.8 icmp_seq=1 ttl=253 time=61.383 ms
84 bytes from 8.8.8.8 icmp_seq=2 ttl=253 time=53.299 ms
84 bytes from 8.8.8.8 icmp_seq=3 ttl=253 time=49.498 ms
8.8.8.8 icmp_seq=4 timeout
8.8.8.8 icmp_seq=5 timeout
8.8.8.8 icmp_seq=6 timeout
8.8.8.8 icmp_seq=4 timeout
8.8.8.8 icmp_seq=5 timeout
8.8.8.8 icmp_seq=6 timeout
8.8.8.8 icmp_seq=4 timeout
8.8.8.8 icmp_seq=5 timeout
84 bytes from 8.8.8.8 icmp_seq=12 ttl=253 time=60.189 ms
84 bytes from 8.8.8.8 icmp_seq=13 ttl=253 time=57.280 ms
84 bytes from 8.8.8.8 icmp_seq=14 ttl=253 time=54.396 ms
84 bytes from 8.8.8.8 icmp_seq=15 ttl=253 time=54.323 ms
84 bytes from 8.8.8.8 icmp_seq=16 ttl=253 time=50.390 ms
84 bytes from 8.8.8.8 icmp_seq=17 ttl=253 time=54.430 ms
84 bytes from 8.8.8.8 icmp_seq=18 ttl=253 time=52.111 ms
84 bytes from 8.8.8.8 icmp_seq=19 ttl=253 time=51.234 ms
84 bytes from 8.8.8.8 icmp_seq=20 ttl=253 time=49.664 ms
```

R1#

```
*Nov 21 20:10:45.087: %TRACKING-5-STATE: 1 ip sla 1 reachability Up->Down
*Nov 21 20:10:48.623: %VRRP-6-STATECHANGE: Fa0/0 Grp 1 state Master -> Backup
```

R1#

R2#

```
*Nov 21 20:10:38.427: %VRRP-6-STATECHANGE: Fa0/0 Grp 1 state Backup -> Master
```

R2#

R1#sh vrrp

FastEthernet0/0 - Group 1

State is Backup

Virtual IP address is 192.168.1.3

Virtual MAC address is 0000.5e00.0101

Advertisement interval is 1.000 sec

Preemption enabled

Priority is 85 (cfgd 105)

Track object 1 state Down decrement 20

Authentication text, string "Ratan"

Master Router is 192.168.1.2, priority is 100
 Master Advertisement interval is 1.000 sec
 Master Down interval is 3.589 sec (expires in 3.141 sec)

R1#

R2#sh vrrp

FastEthernet0/0 - Group 1

State is Master

Virtual IP address is 192.168.1.3

Virtual MAC address is 0000.5e00.0101

Advertisement interval is 1.000 sec

Preemption enabled

Priority is 100

Authentication text, string "Ratan"

Master Router is 192.168.1.2 (local), priority is 100

Master Advertisement interval is 1.000 sec

Master Down interval is 3.609 sec

R2#

R1#sh ip sla summary

IPSLAs Latest Operation Summary

Codes: * active, ^ inactive, ~ pending

ID	Type	Destination	Stats (ms)	Return Code	Last Run
*1	icmp-echo	8.8.8.8	-	Timeout	4 seconds ago

R1#sh track

Track 1

IP SLA 1 reachability

Reachability is Down

3 changes, last change 00:03:07

Latest operation return code: Timeout

Tracked by:

VRRP FastEthernet0/0 1

R1#

R2#sh arp

Protocol	Address	Age (min)	Hardware Addr	Type	Interface
Internet	192.168.1.2	-	ca02.02b4.0008	ARPA	FastEthernet0/0
Internet	192.168.1.3	-	0000.5e00.0101	ARPA	FastEthernet0/0
Internet	192.168.1.10	0	0050.7966.6800	ARPA	FastEthernet0/0
Internet	192.168.1.11	0	0050.7966.6801	ARPA	FastEthernet0/0
Internet	192.168.1.12	0	0050.7966.6802	ARPA	FastEthernet0/0
Internet	200.200.200.1	-	ca02.02b4.0006	ARPA	FastEthernet0/1
Internet	200.200.200.2	71	ca04.1438.0008	ARPA	FastEthernet0/1

R2#

Test – Simulate link recovery by making ISP-1 fa0/1 interface up.

ISP-1#config t

ISP-1(config)#int fa0/1

ISP-1(config-if)#no shut

ISP-1(config-if)#exit

ISP-1#

R1#

*Nov 21 20:16:45.091: %TRACKING-5-STATE: 1 ip sla 1 reachability Down->Up

*Nov 21 20:16:47.703: %VRRP-6-STATECHANGE: Fa0/0 Grp 1 state Backup -> Master

R1#

R2#

*Nov 21 20:16:37.619: %VRRP-6-STATECHANGE: Fa0/0 Grp 1 state Master -> Backup

R2#

R1#sh vrrp

FastEthernet0/0 - Group 1

State is Master

Virtual IP address is 192.168.1.3

Virtual MAC address is 0000.5e00.0101

Advertisement interval is 1.000 sec

Preemption enabled

Priority is 105

Track object 1 state Up decrement 20

Authentication text, string "Ratan"

Master Router is 192.168.1.1 (local), priority is 105

Master Advertisement interval is 1.000 sec

Master Down interval is 3.589 sec

R1#sh track

Track 1

IP SLA 1 reachability

Reachability is Up

4 changes, last change 00:01:15

Latest operation return code: OK

Latest RTT (milliseconds) 72

Tracked by:

VRRP FastEthernet0/0 1

R1#

R2#sh vrrp

FastEthernet0/0 - Group 1

State is Backup

Virtual IP address is 192.168.1.3

Virtual MAC address is 0000.5e00.0101

Advertisement interval is 1.000 sec

Preemption enabled

Priority is 100

Authentication text, string "Ratan"

Master Router is 192.168.1.1, priority is 105
 Master Advertisement interval is 1.000 sec
 Master Down interval is 3.609 sec (expires in 3.325 sec)
 R2#

To Remove VRRP and IP SLA

On R1 router:

```
R1#config t
R1(config)#int fa0/0
R1(config-if)#no vrrp 1
R1(config-if)#exit
R1(config)#no ip sla 1
R1(config)#no track 1
R1(config)#exit
R1#
```

***Nov 21 20:19:45.887: %VRRP-6-STATECHANGE: Fa0/0 Grp 1 state Master -> Disable**

R1#

On R2 router:

```
R2#config t
R2(config)#int fa0/0
R2(config-if)#no vrrp 1
R2(config-if)#exit
R2(config)#exit
R2#
```

***Nov 21 20:19:59.363: %VRRP-6-STATECHANGE: Fa0/0 Grp 1 state Master -> Disable**

R2#

Important Commands:

```
sh ip int brief | exclude unassign  
sh ip route static  
sh ip route ospf  
sh vrrp  
sh vrrp brief  
sh ip nat translations  
sh arp
```

Ratan